

Python SQLite3

Database Connection

Code:

```
import sqlite3

try:
    sqliteConnection = sqlite3.connect('sql.db')
    cursor = sqliteConnection.cursor()
    print('DB Init')

    query = 'SELECT sqlite_version();'
    cursor.execute(query)

    result = cursor.fetchall()
    print('SQLite Version is {}'.format(result[0][0]))

    cursor.close()

except sqlite3.Error as error:
    print('Error occurred -', error)

finally:
    if sqliteConnection:
        sqliteConnection.close()
    print('SQLite Connection closed')
```

Output:

```
DB Init
SQLite Version is 3.50.4
SQLite Connection closed
```

DML Commands:

Code:

```
import sqlite3
```

```
conn = sqlite3.connect('geeks.db')
```

```
cursor = conn.cursor()
```

```
cursor.executescript("""
```

```
DROP TABLE IF EXISTS STUDENT;
```

```
CREATE TABLE STUDENT(
```

```
    NAME VARCHAR(255),
```

```
    CLASS VARCHAR(255),
```

```
    SECTION VARCHAR(255)
```

```
);
```

```
INSERT INTO STUDENT VALUES ('Raju', '7th', 'A');
```

```
INSERT INTO STUDENT VALUES ('Shyam', '8th', 'B');
```

```
INSERT INTO STUDENT VALUES ('Baburao', '9th', 'C');
```

```
""")
```

```
conn.commit()
```

```
print("Data Inserted in the table:")
```

```
cursor.execute("SELECT * FROM STUDENT")
```

```
for row in cursor.fetchall():
```

```
    print(row)
```

```
cursor.execute("""UPDATE STUDENT SET CLASS = '10th' WHERE NAME = 'Raju';""")
```

```
print('\nAfter Updating...\n')
```

```
print("STUDENT Table: ")
```

```
data = cursor.execute("SELECT * FROM STUDENT")
```

```
for row in data:
    print(row)

query = "SELECT * FROM STUDENT where NAME = 'Raju'"
print("Raju Details: ")
cursor.execute(query)
for row in cursor.fetchall():
    print(row)
conn.close()

conn.close()
```

Output:

```
Data Inserted in the table:
('Raju', '7th', 'A')
('Shyam', '8th', 'B')
('Baburao', '9th', 'C')

After Updating...

STUDENT Table:
('Raju', '10th', 'A')
('Shyam', '8th', 'B')
('Baburao', '9th', 'C')
Raju Details:
('Raju', '10th', 'A')
```

Application Using sqlite3 and flask:

Code:

```
CREATE TABLE IF NOT EXISTS users (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    username TEXT UNIQUE NOT NULL,  
    email TEXT UNIQUE NOT NULL,  
    password_hash TEXT NOT NULL,  
    role TEXT NOT NULL CHECK(role IN ('admin', 'member')),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE IF NOT EXISTS books (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    title TEXT NOT NULL,  
    author TEXT NOT NULL,  
    isbn TEXT UNIQUE NOT NULL,  
    genre TEXT,  
    description TEXT,  
    cover_url TEXT,  
    total_copies INTEGER NOT NULL DEFAULT 1,  
    available_copies INTEGER NOT NULL DEFAULT 1,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE IF NOT EXISTS rentals (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    user_id INTEGER NOT NULL,  
    book_id INTEGER NOT NULL,  
    borrow_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    due_date TIMESTAMP NOT NULL,  
    return_date TIMESTAMP,  
    status TEXT NOT NULL CHECK(status IN ('pending', 'approved', 'returned', 'rejected', 'overdue')),
```

```
fine_amount REAL DEFAULT 0.0,  
FOREIGN KEY(user_id) REFERENCES users(id) ON DELETE CASCADE,  
FOREIGN KEY(book_id) REFERENCES books(id) ON DELETE CASCADE  
);
```

Output:

The screenshot shows the 'Aura Library' application interface. The top navigation bar includes 'Dashboard', 'Books', 'Rentals', 'Members', and 'Database'. The 'Database' section is active, displaying the 'SQLite3 Database Viewer'. Below the viewer title, it indicates a 'Live read-only view of db.sqlite' with '3 tables'. The 'users' table is selected, showing 3 rows. The table columns are: ID, USERNAME, EMAIL, PASSWORD_HASH, ROLE, and CREATED_AT. The data rows are: 1 (admin, admin@library.com, admin), 2 (user, user@library.com, member), and 3 (prajith, prajithk24@gmail.com, member).

ID	USERNAME	EMAIL	PASSWORD_HASH	ROLE	CREATED_AT
1	admin	admin@library.com	script:32768-8:1\$3Hx...	admin	2026-07-13 08:24:08
2	user	user@library.com	script:32768-8:1\$Yim...	member	2026-07-13 08:24:08
3	prajith	prajithk24@gmail.com	script:32768-8:1\$0y...	member	2026-07-13 09:05:01

The screenshot shows the 'Aura Library' application interface. The top navigation bar includes 'Dashboard', 'Books', 'Rentals', 'Members', and 'Database'. The 'Database' section is active, displaying the 'SQLite3 Database Viewer'. Below the viewer title, it indicates a 'Live read-only view of db.sqlite' with '3 tables'. The 'books' table is selected, showing 7 rows. The table columns are: ID, TITLE, AUTHOR, ISBN, GENRE, DESCRIPTION, COVER_URL, TOTAL_COPIES, and AVAILABL. The data rows are: 1 (The Great Gatsby, F. Scott Fitzgerald, Fiction, 5), 2 (To Kill a Mockingbird, Harper Lee, Fiction, 3), 3 (1984, George Orwell, Dystopian, 4), 4 (Dune, Frank Herbert, Sci-Fi, 3), 5 (The Hobbit, J.R.R. Tolkien, Fantasy, 6), 6 (Atomic Habits, James Clear, Self-Help, 8), and 7 (Project Hall Mary, Andy Weir, Sci-Fi, 3).

ID	TITLE	AUTHOR	ISBN	GENRE	DESCRIPTION	COVER_URL	TOTAL_COPIES	AVAILABL
1	The Great Gatsby	F. Scott Fitzgerald	9780743273565	Fiction	The novel tells the tragic story of Jay Gatsb...	link	5	5
2	To Kill a Mockingbird	Harper Lee	9780061120084	Fiction	The unforgettable novel of a childhood in a ...	link	3	2
3	1984	George Orwell	9780451524935	Dystopian	Winston Smith reins in his rebellion against t...	link	4	4
4	Dune	Frank Herbert	9780441172719	Sci-Fi	Set in the far future amidst a sprawling feud...	link	3	3
5	The Hobbit	J.R.R. Tolkien	9780261102217	Fantasy	Bilbo Baggins is a hobbit who enjoys a comf...	link	6	6
6	Atomic Habits	James Clear	9780735211292	Self-Help	No matter your goals, Atomic Habits offers ...	link	8	8
7	Project Hall Mary	Andy Weir	9780593135204	Sci-Fi	Ryland Grace is the sole survivor on a desp...	link	3	3



Dashboard

Books

Rentals

Members

Database

Welcome, admin

ADMIN

Logout

 SQLite3 Database Viewer

Live read-only view of db.sqlite • 3 tables

← Back to Dashboard

users (3)

books (7)

rentals (2)

 RENTALS

id, user_id, book_id, borrow_date, due_date, return_date, status, fine_amount, username, book_title

2 rows

ID	USER_ID	BOOK_ID	BORROW_DATE	DUE_DATE	RETURN_DATE	STATUS	FINE_AMOUNT	USERNAME	BOOK_TITLE
2	3	2	2026-07-13 14:35:30	2026-07-27 14:35:30	NULL	approved	\$0.00	prajith	To Kill a Mockingbird
1	2	7	2026-07-13 14:20:57	2026-07-27 14:20:57	2026-07-13 14:33:16	returned	\$0.00	user	Project Hail Mary

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